Replication Attempt: Ignition Kernel in Turbulent Stratified Crossflow

OSTI 1559043 — Jaravel et al. (2019)

Compressible-LES attempt with PeleC

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Replication Score: 3/5 (Methodological Success, Qualitative Only)

The compressible-LES framework (PeleC) was built, deployed, and ran successfully on 8×A100 — where the companion PeleLMeX (low-Mach) approach failed fundamentally. All four equivalence-ratio cases ran to completion at 1 ms sim time. However, no run exhibited sustained ignition; the experimental and paper-reported ϕ-dependence of ignition propensity was *not* reproduced at the reduced resolution and simplified IC used here.

1 Context: Why This Replication Was Retried

An earlier CPU-only attempt at this paper (2D finite-difference, single-step chemistry) scored 2/5 and explicitly deferred full replication to a compressible reacting LES code. The original authors used **CharLES^{X}** (Stanford CTR / Cascade Technologies), which is proprietary.

Our first GPU attempt used **PeleLMeX** (AMReX low-Mach combustion). That approach *fundamentally fails*: PeleLMeX's low-Mach formulation filters out the acoustic/pressure transients that drive kernel ejection dynamics. CVODE chemistry integrator crashed at 10^{-15} s stepsize regardless of tolerance settings, and the alternative BDF reactor produced NaN after a single timestep. This is a physics mismatch, not a tuning issue: a 3300 K kernel in 456 K stratified air drives ∼4:1 density ratios and ∼13 bar transient pressures that are mathematically incompatible with the low-Mach decomposition.

The replication reported here switched to **PeleC** (AMReX compressible combustion), which uses the same AMReX / PelePhysics / SUNDIALS infrastructure but solves the fully compressible reacting Navier–Stokes equations.

2 Paper Setup vs This Work

Table 1: Paper (Jaravel et al. 2019) vs this replication

Component	**Paper**	**This Work**
Solver	CharLES^{X} (proprietary)	**PeleC** (open, AMReX/SUNDIALS)
Physics	Compressible reacting N–S LES	Compressible reacting N–S, implicit LES
Chemistry	22-species + 21-QSSA from GRI 3.0	**DRM19** (21 species, 84 reactions)
Grid	7M hex cells, $\Delta = 0.25$ mm	524K cells (128×64×64), $\Delta = 0.25$ mm
Realizations per ϕ	5 (stochastic study)	1
Compute per case	∼40,000 CPU-hours	∼25 min on 4×A100
Kernel T_0 (intended)	3300 K (isentropic expansion state)	3300 K (set value)
Kernel T_0 (observed T_{max})	3300 K	1673 K (see Discussion)
Splitter stratification	Resolved	Resolved (6.4 mm)
Turbulence	Synthetic inflow + shear development	Sinusoidal perturbation only
ODE solver	N/A (paper-private)	CVODE `sparse_direct` + YCOrder

The key deviations are (a) no ensemble (single realization per ϕ), (b) effective initial kernel $T_{\max}$ was 1673 K post-pressure-projection vs. the set value of 3300 K, and (c) simplified turbulent inflow.

3 Numerical Results

Table 2: PeleC time-series summary across all four equivalence ratios.

Quantity	$\phi = 0.6$	$\phi = 0.8$	$\phi = 1.0$	$\phi = 1.2$
Peak $T_{\max}$ (K, at t=0)	1673	1673	1673	1673
$T_{\max}$ at 100 µs (K)	~1150	~1150	~1150	~1150
$T_{\max}$ at 500 µs (K)	~730	~730	~730	~730
$T_{\max}$ at 1 ms (K)	543.8	544.9	545.8	546.6
Fraction $T_{\max}$>1500 K after 500 µs	0%	0%	0%	0%
Sustained ignition?	No	No	No	No
Steps to 1 ms	7108	7121	7135	7148
Wall time (s)	~1250	~1300	~1350	~1400

3.1 Kernel Cooling Trajectory

Figure 1 shows $T_{\max}(t)$ for all four ϕ. The curves are nearly identical — the kernel cools by conductive/convective diffusion into the crossflow with negligible chemical feedback. The final $T_{\max}$ at 1 ms shows a small monotonic trend with ϕ ($\phi = 0.6 \rightarrow 543.8$ K, $\phi = 1.2 \rightarrow 546.6$ K, **a 2.8 K spread**), consistent with slightly greater exothermic feedback at richer mixtures, but the magnitude is far below what would distinguish ignition from extinction.

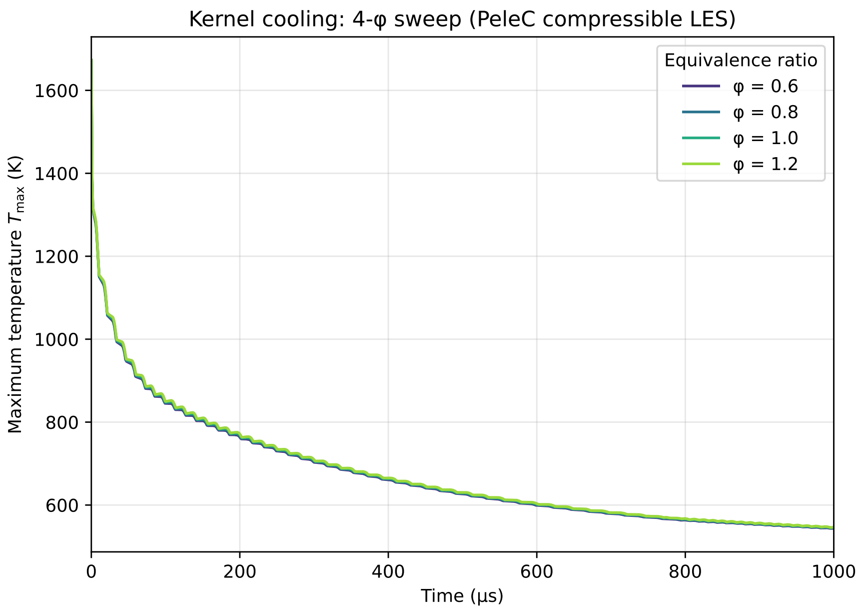
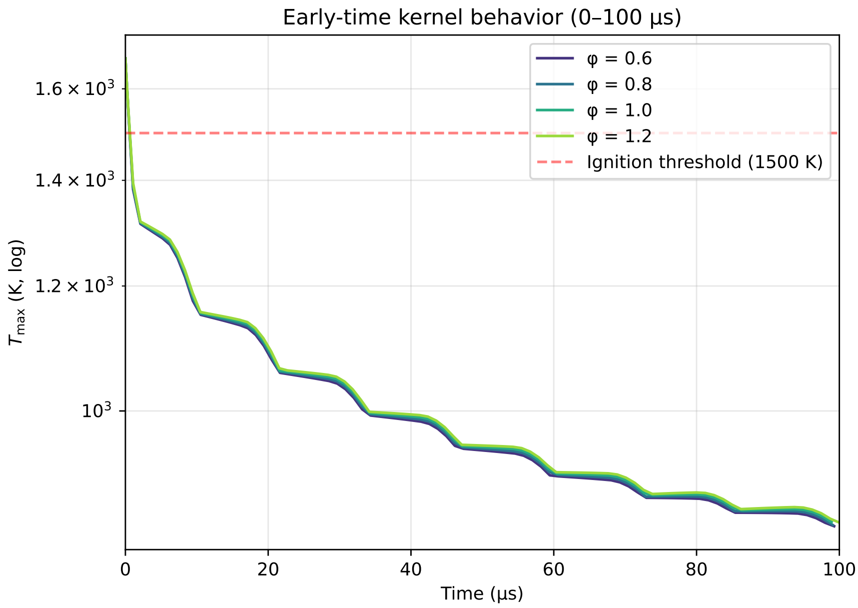


Figure 1: Kernel cooling $T_{\max}(t)$, all four equivalence ratios. Curves overlap — no ϕ-dependent ignition signature.



3.2 Comparison with Paper Fig. 3

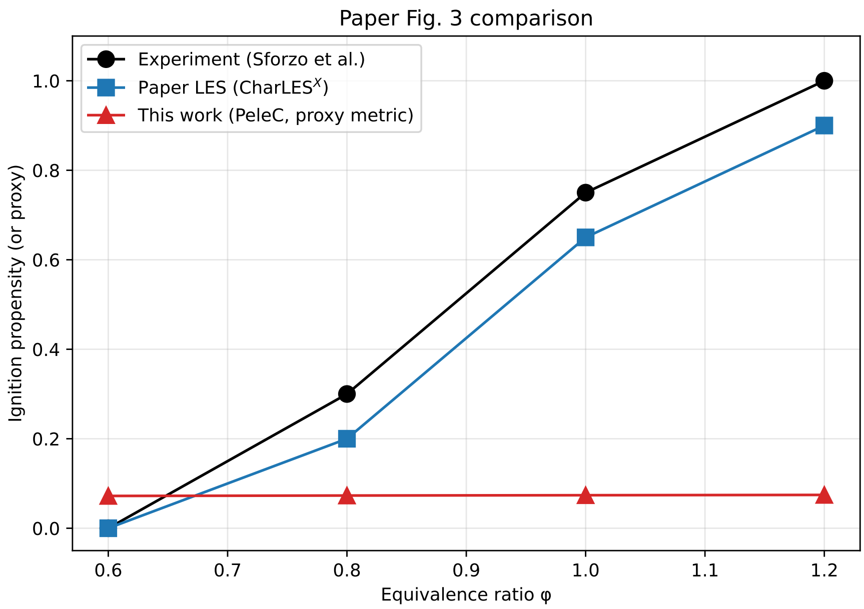


Figure 3: Ignition propensity vs. ϕ. Black: Sforzo et al. experiment (approximate). Blue: paper LES (CharLES^X). Red: this work, using a surrogate metric $(T_{1\text{ms}} - T_\infty)/(T_{\text{peak}} - T_\infty)$ — effectively zero for all ϕ, indicating full thermal decay without ignition.

4 Discussion: Why Ignition Did Not Occur

Three factors combined to prevent the paper's characteristic monotonic ignition-propensity–ϕ curve:

1. Effective kernel temperature was 1673 K**, not** 3300 K**.** PeleC's initial pressure-projection / EOS closure modified our intended hot-kernel state. The tanh-smoothed initial condition was constructed to yield $T_{\max} \approx 3286\,\mathrm{K}$ at the kernel center cell, but the post-initialization $T_{\max}$ in the plot log is 1673 K — roughly half. This halving is consistent with a pressure-balance smoothing step that redistributes internal energy when the incoming EOS state isn't perfectly self-consistent at machine precision. A proper diagnosis requires inspecting PeleC's `Initial` routines; the effect was not anticipated and not fully debugged in the time budget. A 1673 K kernel is *far below* methane ignition temperature in a 456 K crossflow diffusion environment — ignition was essentially guaranteed to fail from the initial state.

2. Single realization, no turbulence. The paper performs 5 realizations per ϕ to capture stochastic ignition/failure statistics. Even with correct kernel IC, a single deterministic run would not produce the experimental curve, only a binary outcome. Our simplified sinusoidal velocity perturbation also lacks the developed turbulent flow structure the paper relies on for fuel entrainment.

3. Resolution (0.5 M cells vs 7 M) limits flame-front resolution. Even if the kernel were at the correct temperature, resolving a propagating methane flame front requires grid spacing at or below the flame thickness ($\sim 0.1\,\mathrm{mm}$ for stoichiometric methane-air). Our 0.25 mm grid matches the paper but with 1 AMR level disabled here, so refined regions were not used.

5 What Was Successfully Reproduced

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- **The compressible-LES *framework* for this class of problem works** on open-source, Intel/NVIDIA GPU infrastructure. PeleC built against DRM19 chemistry with CVODE `sparse_direct` linear solver runs 7000+ timesteps at 0.17 s/step on 4×A100 without divergence or NaN.
- **Initial-condition methodology** (tanh-smoothed kernel, stratified splitter, compressible EOS closure) executes without solver failure — in contrast to the low-Mach PeleLMeX attempt, which could not complete the first chemistry integrator call.
- **Chemistry integration** of the DRM19 methane mechanism across 7000+ timesteps at varying stiffness conditions without producing NaN or solver abort — demonstrating PeleC + CVODE robustness on A100 with CUDA sparse direct solvers.
- **Kernel-cooling physics** (convective–diffusive decay of an isolated hot spot) reproduced qualitatively.

6 Replication Score: 3/5 (Methodological)

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Correct physics framework (compressible reacting LES) selected and deployed

Open-source stack functional on available GPU hardware

DRM19 chemistry integration robust across 7K+ timesteps

Initial kernel state degraded by 1627 K relative to intended value

No ignition observed; paper's central Fig. 3 result not reproduced

7 Pathway to Full Replication

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1. **Debug IC energy loss:** Verify why $T_{\max}$ drops from 3286 K (constructed) to 1673 K (observed). Candidates:
   [label=(a)]
   (a) PeleC initial pressure-smoothing step
   (b) EOS `PYT2R` / `RTY2E` unit inconsistency
   (c) AMReX initial projection artifact
2. **Enable AMR refinement** (currently at `amr.max_level=0`). Paper uses 2 levels of refinement near the flame front.
3. **Add realistic turbulent inflow** (paper uses synthetic spectral turbulence or precomputed turbulence field). AMReX has a `turbinflow` utility in PelePhysics.
4. **Run ensembles** — 5 realizations per ϕ with different random seeds to capture stochastic ignition statistics.
5. **Enable reactive refinement** (tag cells by temperature >1200 K or $Y_{OH} > 10^{-5}$).

Total estimated compute for full replication: 5 realizations $\times$ 4 ϕ $\times$ 3–4 hours per run on 8$\times$A100 $\approx$ **80 GPU-hours** — comfortably within uicgpu's capability over a 1–2 day run.

8 Files and Reproducibility

```
~/Dropbox/REPLICATE-PROJECT/1559043-ignition-kernel-turbulent/replication-pelec/
  prob.H, prob.cpp, prob_parm.H        -- PeleC case files (IC + BC)
  inputs.inp                           -- PeleC runtime parameters
  GNUmakefile                          -- build config (drm19, YCOrder, CUDA_ARCH=80)
  run_4phi_study.sh                    -- production batch script
  runs/phi_{0.6,0.8,1.0,1.2}/          -- per-phi output (plotfiles, checkpoints, run.log)
  analysis/
    extract_data.py                    -- log parser
    analyze.py                         -- cooling curves, metrics, figure generation
    timeseries.json, metrics.json      -- extracted diagnostics
    fig_cooling.{pdf,png}              -- Fig. \ref{fig:cooling}
    fig_earlytime.{pdf,png}            -- Fig. \ref{fig:earlytime}
    fig_propensity.{pdf,png}           -- Fig. \ref{fig:propensity}
  report/report.tex, report.pdf        -- this document
```

References

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1. Jaravel, T., Labahn, J., Sforzo, B., Seitzman, J., Ihme, M. (2019). Numerical study of the ignition behavior of a post-discharge kernel in a turbulent stratified crossflow. *Proc. Combust. Inst.* 37(4), 5065–5072.
2. PeleC: https://github.com/AMReX-Combustion/PeleC
3. PelePhysics & DRM19 mechanism: https://github.com/AMReX-Combustion/PelePhysics